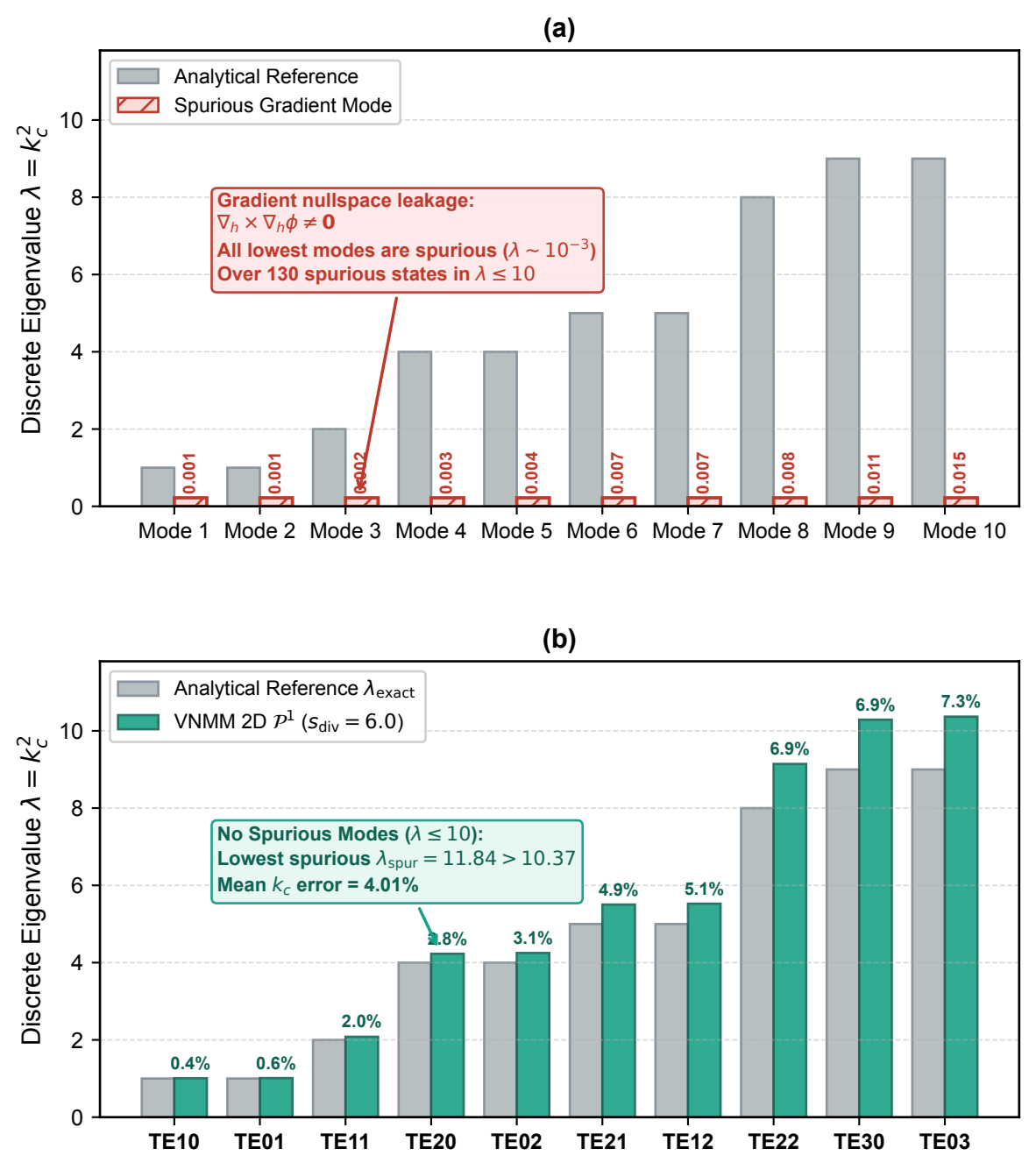

(a)
Analytical Reference
Spurious Gradient Mode
Gradient nullspace leakage:
$\nabla_h \times \nabla_h \phi \neq \mathbf{0}$
All lowest modes are spurious ($\lambda \sim 10^{-3}$)
Over 130 spurious states in $\lambda \leq 10$
Discrete Eigenvalue $\lambda = k_c^2$
0.001
0.001
0.002
0.003
0.004
0.007
0.007
0.008
0.011
0.015
Mode 1
Mode 2
Mode 3
Mode 4
Mode 5
Mode 6
Mode 7
Mode 8
Mode 9
Mode 10
(b)
Analytical Reference λ_{exact}
VNMM 2D $\mathcal{P}^1$ ($s_{\text{div}} = 6.0$)
No Spurious Modes ($\lambda \leq 10$):
Lowest spurious $\lambda_{\text{spur}} = 11.84 > 10.37$
Mean k_c error = 4.01%
Discrete Eigenvalue $\lambda = k_c^2$
0.4%
0.6%
2.0%
3.8%
3.1%
4.9%
5.1%
6.9%
6.9%
7.3%
TE10
TE01
TE11
TE20
TE02
TE21
TE12
TE22
TE30
TE03